THE COMPLEXITY OF SIGNED H-COLOURING

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Abstract. Verifying a conjecture of Brewster, Foucaud, Hell and Naserasr, we show that signed (H, Π)-colouring is NP-complete for any signed graph (H, Π) whose s-core has at least 3 edges.

1. Introduction

The celebrated Constraint Satisfaction Problem (CSP) Dichotomy Conjecture of Feder and Vardi [9], with recently announced proofs [5, 6, 16], demonstrates the fundamental role graph homomorphisms and their variants play in the study of computational complexity. The general set-up is the following decision problem. Let $\mathcal{H}$ be a fixed relational structure.

Hom($\mathcal{H}$)
Instance: A relational structure $\mathcal{G}$ over the same vocabulary as $\mathcal{H}$.
Question: Does $\mathcal{G}$ admit a homomorphism to $\mathcal{H}$?

The CSP Dichotomy Conjecture asserts Hom($\mathcal{H}$) is either NP-complete or polynomial time solvable for each $\mathcal{H}$. In the special case of graph homomorphisms, Hell and Nešetřil [11] give a particularly simple dichotomy classification: if the core of H (defined below) is a graph on at most one edge (a vertex, an edge, or a loop), the homomorphism problem Hom(H) is polynomial; otherwise, it is NP-complete. Other homomorphism problems with easily described dichotomies include homomorphisms to smooth digraphs [2] and list-homomorphisms [7]. By contrast, we have no concise structural characterization of the dichotomy for all of CSP; for this one must delve into notions of universal algebra.

In [3], the authors study the complexity of signed graph homomorphisms. They propose a simply stated dichotomy along the lines of Hell and Nešetřil. If a signed graph (H, Π) has, or reduces as described below to a signed graph that has, at most two edges, then the homomorphism problem s-Hom(H, Π) is polynomial; otherwise the problem is NP-complete. In [3], they prove the polynomial side of the conjecture and establish all the NP-complete cases with one exception. In this note we settle the final case and thereby prove the proposed dichotomy.

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2. Defintions and Tools

A *signed graph* is a graph G together with a signing function $\sigma : E(G) \to \{+,-\}$. By setting $\Sigma = \sigma^{-1}(-)$, we use the notation (G,Σ) to denote this signed graph. We allow loops and parallel edges in G. In the case there are parallel edges between a pair of vertices, we may naturally assume there are only two edges — one signed with + and the other with −.

Denoting the set of positive edges by $E_1(G)$ and the set Σ of negative edges by $E_2(G)$, a signed graph (G,Σ) may also be viewed as a 2-edge-coloured graph, i.e. a relational system with two binary, symmetric relations on the same vertex set $V(G)$. However, fundamental to the study of signed graphs is the fact that $\{+,-\}$ forms a group under multiplication. Hence any subset of edges may be assigned a value of *positive* or *negative* by multiplying the signs of all the edges in the set. In particular, a cycle is *balanced* if its set of edges is positive and *unbalanced* otherwise.

A second important notion for signed graphs is the operation of *switching*, introduced by Zaslavsky [18]. To *switch* at a vertex v means to multiply the signs of all edges incident to v by −, that is, to switch the sign of each of these edges. (In the case of a loop at v, its sign is multiplied twice and hence it is invariant under switching.) If (G,Σ') is obtained from (G,Σ) by a series of switches, then we say (G,Σ) and (G,Σ') are *switch equivalent*. Switch equivalence is an equivalence relation. Since the process of switching cannot change the balance of a cycle, equivalent signed graphs necessarily have the same balance on all cycles. In fact, this condition is also sufficient for signed graphs to be equivalent [18].

Our study of homomorphisms of signed graphs flows from the paper [14]. The first complexity results appear in [10]. The concepts of ec-homomorphisms and s-homomorphisms below appear in [3, 15].

Let (G,Σ) and (H,Π) be signed graphs. An *ec-homomorphism* of (G,Σ) to (H,Π), denoted $\varphi : (G,\Sigma) \xrightarrow{ec} (H,\Pi)$ is a mapping $\varphi : V(G) \to V(H)$ such that $xy \in E_i(G)$ implies $\varphi(x)\varphi(y) \in E_i(H)$. Thus an ec-homomorphism is a graph homomorphism of G to H that preserves edge colours. An *s-homomorphism* of (G,Σ) to (H,Π), denoted $\varphi : (G,\Sigma) \xrightarrow{s} (H,\Pi)$, is a resigning of (G,Σ) to (G,Σ') and of (H,Π) to (H,Π') together with an ec-homomorphism $\varphi : (G,\Sigma') \xrightarrow{ec} (H,\Pi')$. It is easy to see that switching in (H,Π) is not necessary, as one can equivalently switch the pre-image in (G,Σ). Hence, when studying s-homomorphisms, we are free to switch (H,Π) to an equivalent signed graph of our choice.

We write $(G,\Sigma) \xrightarrow{ec} (H,\Pi)$ to indicate the existence of a ec-homomorphism when the name of the map is clear or unimportant. Similarly, we may write $(G,\Sigma) \xrightarrow{s} (H,\Pi)$. Alternatively one can define ec-homomorphisms and s-homomorphisms simply as a graph homomorphism $G \to H$ that preserves edge-colours for the former and cycle balance for the latter. See [3].

The focus of [3] and this paper are two special cases of $\textsc{Hom}(\mathcal{H})$ for signed graphs. Let (H,Π) be a fixed signed graph.

ec-$\textsc{Hom}(H,\Pi)$
Instance: A signed graph (G,Σ).
Question: Does $(G,\Sigma) \xrightarrow{ec} (H,\Pi)$?

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s-Hom(H,Π)
Instance: A signed graph (G,Σ).
Question: Does $(G,\Sigma) \xrightarrow{s} (H,\Pi)$?

In [3] the authors exploit the following well known connection between ec-homomorphisms and s-homomorphisms. See also [4, 13, 15].

Definition 2.1. Let (H,Π) be a signed graph. The *switching graph* of (H,Π) is a signed graph denoted $P(H,\Pi)$ and constructed as follows.

(i) For each vertex u in $V(H)$ we have two vertices u_0 and u_1 in $P(H,\Pi)$.
(ii) For each edge e between u and v in H, we have four edges between u_i and v_j $(i,j \in \{0,1\})$ in $P(H,\Pi)$, with the edges between u_i and v_i having the same sign as e and the edges between u_i and v_{1-i} having sign opposite to that of e $(i \in \{0,1\})$.

See Figure 1 for an example of the switching graph construction. We use blue, solid edges for positive edges and red, dashed edges for negative edges. Key to our work is the following observation.

Proposition 2.2 ([3])**.** *Let (G,Σ) and (H,Π) be signed graphs. Then $(G,\Sigma) \xrightarrow{s} (H,\Pi)$ if and only if $(G,\Sigma) \xrightarrow{ec} P(H,\Pi)$.*

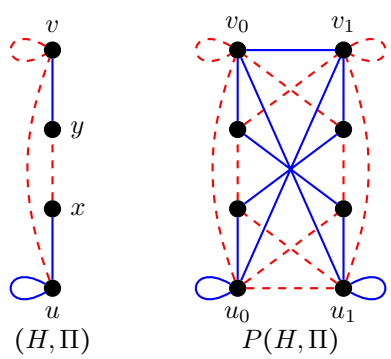

Figure 1. A signed graph (H,Π) and the switching graph $P(H,\Pi)$.

The following ideas are basic in the field of graph colouring and CSP problems. Details can be found in many places; see, for example, [12]. We recall a relational structure $\mathcal{H}$, and hence a 2-edge-coloured graph, is a *core* if its only endomorphisms are automorphisms. Up to automorphism of $\mathcal{H}$, there is a unique smallest endomorphic image in $\mathcal{H}$. This is necessarily a core, and is called *the core of* $\mathcal{H}$. We extend this notion to signed graphs and say (H,Π) is an *s-core* if every s-homomorphism $\varphi : (H,\Pi) \xrightarrow{s} (H,\Pi)$ is a bijection. As $\mathcal{H}$ and its core are homomorphically equivalent, it is standard to restrict the study of Hom$(\mathcal{H})$ to the case that $\mathcal{H}$ is a core. Similarly, we restrict the study of s-Hom(H,Π) to the case that (H,Π) is an s-core.

A 2-edge-coloured graph is just an example of a relational structure, so by several recent results [5, 6, 16], we know that that the problem ec-Hom(H,Π) is either polynomial time solvable or NP-complete, for any signed graph (H,Π). These results however, do not tell us, in an easily computable way, for which (H,Π) it is

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NP-complete. In fact, in [3], they show that for any relational structure $\mathcal{H}$ there is a bipartite signed graph H such that $\textsc{Hom}(\mathcal{H})$ and $\textsc{ec-Hom}(H,\Pi)$ are polynomially equivalent. On the other hand, the following is Conjecture 1.6 of [3] which asserts that the dichotomy for $\textsc{s-Hom}(H,\Pi)$ is easily described.

Conjecture 2.3 ([3])**.** *For a connected s-core (H,Π), $\textsc{s-Hom}(H,\Pi)$ is polynomially time solvable if (H,Π) has at most two edges, and is otherwise NP-complete.*

The authors of [3] verified the polynomial side of the conjecture, and on the other side, showed that $\textsc{s-Hom}(H,\Pi)$ is NP-complete for any s-core with at least 3 edges not having a loop of each sign and a *digon*: a pair of vertices with both a positive and negative edge between them.

In this short note, we verify that the conjecture is true. It follows immediately from Theorem 4.3.

3. Disconnected loops

In our proof, we use the following two main results. The first is a seminal result of Hell and Nešetřil.

Theorem 3.1 ([11])**.** *Let H be a graph that is a core. If H has at most one edge, then $\textsc{Hom}(H)$ is polynomial time solvable; otherwise, H is loop-free and contains an odd cycle, in which case, $\textsc{Hom}(H)$ is NP-complete.*

The second result is from [8] where they study the *retraction problem* $\textsc{Ret}(\mathrm{H})$ for graphs H containing some looped vertices. An instance of $\textsc{Ret}(H)$ is a graph G containing a labelled subgraph isomorphic to H. The graph G retracts to H if $\varphi : G \to H$ where φ is the identity on labelled subgraph H.

Theorem 3.2. [8] *Let H be a connected graph in which loops are allowed. If the subgraph of H induced by vertices with loops is not connected, then $\textsc{Ret}(H)$ is NP-complete.*

In our work on Conjecture 2.3 we independently found a nice proof of Theorem 3.2. We include it here as it may be of independent interest. The set-up of [8] and our proof is similar, but the tools and approaches are quite different.

A *polymorphism* of a relational structure $\mathcal{H}$ is a homomorpshim $\phi : \mathcal{H}^k \to \mathcal{H}$. (The product here is the categorical or tensor product.) A polymorphism ϕ is *idempotent* if for any vertex $v \in \mathcal{H}$ we have $\phi(v,v,\ldots,v) = v$. Several people have observed that the following theorem follows from [2] by the proof of [17]. This first occurs in print in [1].

Theorem 3.3. *If a core relational structure $\mathcal{H}$ admits no idempotent polymorphism $\sigma : \mathcal{H}^4 \to \mathcal{H}$ satisfying the identity*

$$\sigma(a,b,c,a) = \sigma(b,a,b,c) \tag{1}$$

then $\mathrm{Hom}(\mathcal{H})$ *is NP-complete.*

We are interested in applying this result to graphs that are not necessarily cores. However, it is well known, as noted in [8], that for a graph H, the relational structure $\mathcal{H}^c$ obtained from $\mathcal{H} = (V(H), E(H))$ by adding a constant relation for each vertex in H has the property that $\textsc{Hom}(\mathcal{H}^c)$ and $\textsc{Ret}(H)$ are polynomially equivalent. In the proof below we prove the non-existence of the polymorphism σ on the graph H, but by considering equivalent core $\mathcal{H}^c$ we can apply Theorem 3.3.

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We use the following notation in the proof: for any vertex v in H and any integer $d \geq 1$, $N^{(d)}(v)$ is the set of all vertices that one can reach from v by a walk of length d.

Proof of Theorem 3.2. Let H be a connected graph whose subgraph L of looped vertices is disconnected. Let d be the length of the shortest path in H between components of L.

Consider a path $P = v_0 \sim v_1 \sim \dots v_{d-1} \sim v_d$ between two components of L, say S_0 and S_d. Let $S_{d-1} = N^{(d-1)}(v_0) \cap N^{(1)}(v_d)$. Let k be the size of the maximum clique in S_{d-1}. Without loss of generality assume that $v_0 \in S_0$ and $v_d \in S_d$ are chosen to maximise k. Let $x_1, x_2, \dots, x_k$ be the vertices of a maximum clique in S_{d-1}.

By Theorem 3.3, it is enough to show that H admits no idempotent polymorphism σ satisfying (1). Towards contradiction, assume that it admits such a polymorphism σ.

It is quite easy to see that $\sigma(v_0v_1v_0v_0)$ is in S_0. Indeed $\sigma(v_0v_1v_0v_0) = \sigma(v_1v_0v_0v_1)$ and $v_0v_1v_0v_0 \sim v_1v_0v_0v_1$ so σ maps $v_0v_1v_0v_0$ and $v_1v_0v_0v_1$ to a loop. As $N(v_0) \supseteq \{v_0, v_1\}$, $\sigma(\mathbf{u}) \sim \sigma(v_0v_0v_0v_0) = v_0$ for any $\mathbf{u} \in \{v_0, v_1\}^4$; hence, the loop is in S_0. Similarly we get $\sigma(v_dv_dv_{d-1}v_d) = \sigma(v_dv_dv_dv_{d-1}) \in S_d$.

Consider the following path in H^4:

$$v_0v_1v_0v_0 \sim v_0v_2v_0v_0 \sim \dots \sim v_0v_dv_0v_0$$

As $v_0v_dv_0v_0$, has a loop, it must map to a looped vertex. However, it is only distance $d-1$ from $v_0v_1v_0v_0$. Hence by the choice of d, $s_0 := \sigma(v_0v_dv_0v_0) = \sigma(v_dv_0v_0v_d) \in S_0$ (using (1)). By a similar argument $s_d := \sigma(v_dv_dv_0v_d) \in S_d$. The path

$$v_0v_dv_0v_0 \sim v_1v_dv_0v_1 \sim \dots \sim v_dv_dv_0v_d$$

shows s_0 and s_d are exactly distance d apart.

We now show that where $X = \{v_dx_iv_0v_d : i = 1, 2, \dots, k\} \cup \{x_1v_dv_0x_1\}$, $\sigma(X)$ is a clique of size $k+1$ in S_{d-1}, contradicing the definition k. As X is indeed a clique of size $k+1$, and S_{d-1} has no loops, it is enough to show that each vertex of X maps to S_{d-1}. We do this by providing, for each vertex x of X a path of length d from $v_0v_dv_0v_0$ or $v_dv_0v_0v_d$ (mapping to s_0), to $v_dv_dv_0v_d$ (mapping to s_d), in which x is the $d-1^{st}$ vertex. Where $(v_0 = u_0^i) \sim u_1^i \sim \dots \sim u_{d-2}^i \sim (u_{d-1}^i = x_i)$ is a path from v_0 to x_i, the path

$$v_dv_0v_0v_d \sim v_du_1^iv_0v_d \sim v_du_2^iv_0v_d \sim \dots \sim v_du_{d-2}^iv_0v_d \sim v_dx_iv_0v_d \sim v_dv_dv_0v_d$$

does this for $v_dx_iv_0v_d \in X$. For the last vertex $x_1v_dv_0x_1$ of X we use the path

$$v_0v_dv_0v_0 \sim u_1^1v_dv_0u_1^1 \sim u_2^1v_dv_0u_2^1 \sim \dots \sim u_{d-2}^1v_dv_0u_{d-2}^1 \sim x_1v_dv_0x_1 \sim v_dv_dv_0v_d.$$

□

4. The proof of Conjecture 2.3

For a signed graph (H, Π) let $B_{(H,\Pi)}$ be the *subgraph* of positive edges of H, i.e. $B_{(H,\Pi)} = (V(H), E(H) \backslash \Pi)$. We call this *blue subgraph* of H (since we use blue to denote positive edges). The signed graph (H, Π) is *blue connected* if $B_{(H,\Pi)}$ is connected. The *red subgraph* $R_{(H,\Pi)} := (V(H), \Pi)$ is similarly defined as the negative edges of (H, Π).

Claim 4.1. *Let (H, Π) be connected. Then $P(H, \Pi)$ is blue connected if and only if (H, Π) contains an unbalanced cycle.*

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Proof. As H is connected, it contains a spanning tree T. We may switch so that T has only positive edges in (H,Π). By construction $P(H,\Pi)$ contains two copies of T, say T_0 and T_1, the first on vertices of the form u_0 and the second on u_1. Thus $B_{P(H,\Pi)}$ has at most two components.

If $P(H,\Pi)$ is blue connected, then there is a positive edge u_0v_1 in $P(H,\Pi)$ joining T_0 and T_1. The corresponding edge u_0v_0 is negative. The fundamental cycle in $T_0 + u_0v_0$ is unbalanced. On the other hand, if $P(H,\Pi)$ is not blue connected, then all edges of the form u_0v_1 in $P(H,\Pi)$ are negative. All edges in (H,Π) are positive and all cycles are balanced. □

Claim 4.2. *Let (H,Π) be a connected signed graph where each vertex has a positive loop. If $P(H,\Pi)$ is blue connected, then $R_{P(H,\Pi)}$ contains an odd cycle.*

Proof. Assume $P(H,\Pi)$ is blue connected. By the previous claim, (H,Π) contains an unbalanced cycle $C := v^0 \sim v^1 \sim \cdots \sim v^n \sim v^0$. We can switch (H,Π) so that all edges of C are positive with the exception of $v^n \sim v^0$ which must be negative. Then $v_0^0 \sim v_1^1 \sim v_0^1 \sim v_1^2 \sim v_0^2 \sim \ldots v_1^n \sim v_0^n \sim v_0^0$ is the required odd cycle in $R_{P(H,\Pi)}$. □

Theorem 4.3. *If (H,Π) is a connected s-core on at least 3 edges containing a positive loop and an unbalanced cycle, then* s-Hom(H,Π) *is NP-complete.*

Proof. By Proposition 2.2, it is enough to show ec-Hom$(P(H,\Pi))$ is NP-complete.

Indeed, by the assumptions, we have by Claim 4.1 that $P(H,\Pi)$ is blue connected. Consider then the subgraph (L,Λ) of (H,Π) induced by vertices with positive loops. It is clear that $P(L,\Lambda)$ is the subgraph of $P(H,\Pi)$ induced by vertices with positive loops.

If $P(L,\Lambda)$ is not connected, then $B_{P(H,\Pi)}$ is connected but its reflexive subgraph is not. Hence Ret$(B_{P(H,\Pi)})$ is NP-complete by Theorem 3.2. Let G be an instance of Ret$(B_{P(H,\Pi)})$ containing a labelled subgraph isomorphic to $B_{P(H,\Pi)}$. Let (G',Σ) be the signed graph obtained from G by signing all edges positive and then adding the negative edges of $P(H,\Pi)$ to the labelled copy of $B_{P(H,\Pi)}$ in G. Let $\varphi : G \to B_{P(H,\Pi)}$ be a retraction. We claim $\varphi : V(G') \to V(H)$ defines an ec-homomorphism. Positive edges are preserved by definition of φ, and negative edges are preserved since φ is the identity on $P(H,\Pi)$.

Since (H,Π) is an s-core, by [4] we may assume $P(H,\Pi)$ is an ec-core. Thus for $\psi : (G',\Sigma) \xrightarrow{ec} P(H,\Pi)$, we have ψ acts as an automorphism, say τ, on the copy of $P(H,\Pi)$ in (G',Σ). Hence $\tau^{-1} \circ \psi$ is the identity on $P(H,\Pi)$. This defines a retraction of G to $B_{P(H,\Pi)}$. This reduction from Ret$(B_{P(H,\Pi)})$ shows ec-Hom$(P(H,\Pi))$ is NP-complete, as required.

If $P(L,\Lambda)$ is connected, then by Claim 4.2 $R_{P(L,\Lambda)}$ contains an odd cycle. Moreover it can have no negative loops as every vertex has a positive loop, and the s-core of (H,Π) is not simply a vertex with a positive and a negative loop. By Theorem 3.1, Hom$(R_{P(L,\Lambda)})$ is NP-complete. Let G be an instance of Hom$(R_{P(L,\Lambda)})$. Obtain the signed graph (G,Σ) by setting $\Sigma = E(G)$ and adding a positive loop to each vertex. Then $G \to R_{P(L,\Lambda)}$ if and only if $(G,\Sigma) \xrightarrow{ec} P(L,\Lambda)$ if and only if $(G,\Sigma) \xrightarrow{ec} P(H,\Pi)$. This reduction shows ec-Hom$(P(H,\Pi))$ is NP-complete, as required. □

As noted after its statement, Conjecture 2.3 was proved except in the case that (H,Π) has a positive loop, a negative loop, and a digon. As both negative loops and

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digons are unbalanced cycles, Theorem 4.3 deals handily with these cases. Thus we have the following.

Corollary 4.4. *Conjecture 2.3 is true.*

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